

Introduction to Computer Forensics

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Criminalistics

- Fancy term for Forensic Science

Forensic Science

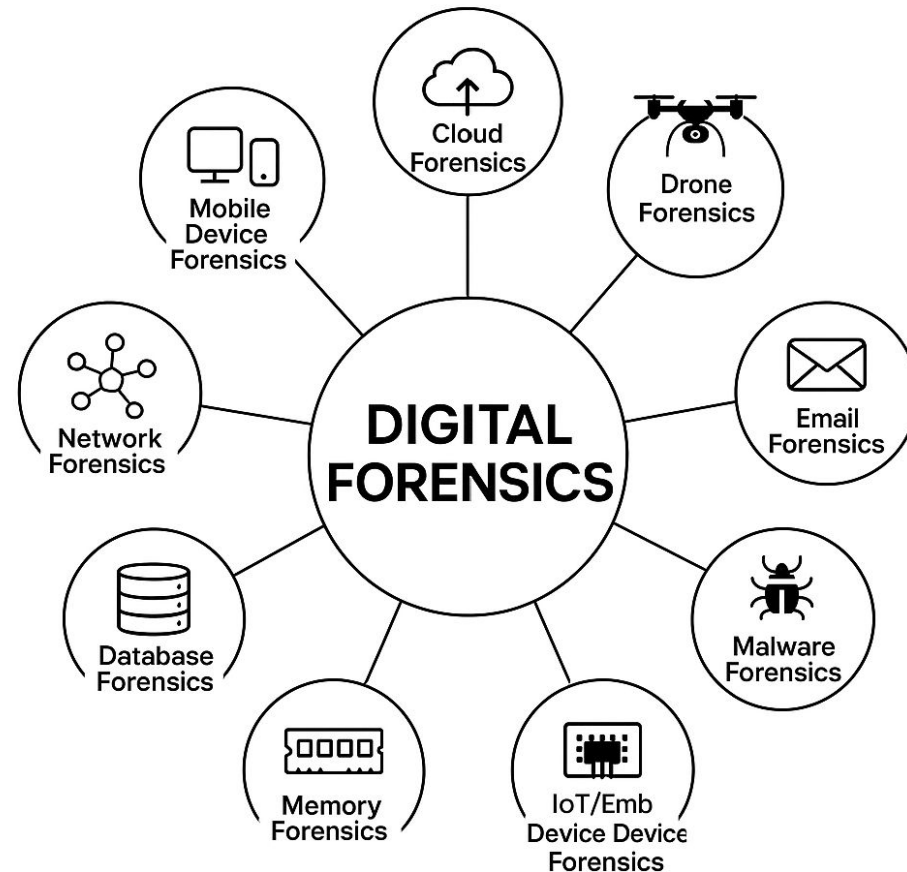
The application of science to those criminal and civil laws that are enforced by police agencies in a criminal justice system (Saferstein, 2004)

Digital Forensics

“Recovering, investigating, and analyzing data from digital devices—such as computers, smartphones, networks, or cloud systems—to uncover evidence of a crime or security incident.”

Branches of Digital Forensics

- Digital forensics is divided into multiple branches based on sources of evidence and technology.



1. Computer Forensics

- Focuses on computers, hard drives, operating systems, and file systems.
- Tools: EnCase, FTK, Autopsy.

2. Mobile Device Forensics

- Deals with smartphones, SIM cards, memory cards.
- Extracts call logs, chat data, GPS, images.
- Tools: Cellebrite, Oxygen Forensics.

3. Network Forensics

- Monitors and analyzes network traffic to detect intrusions and data breaches.
- Tools: Wireshark, Zeek (Bro), Snort.

4. Cloud Forensics

- Focuses on forensic investigation of cloud platforms like AWS, Azure, Google Cloud.
- Challenges: multi-tenancy, remote storage, jurisdiction.

5. Memory (RAM) Forensics

- Involves analyzing volatile memory (RAM) to find malware, hidden processes.
- Tools: Volatility, Rekall.

6. Database Forensics

- Examines data stored in databases like MySQL, Oracle.
- Finds logs, transactions, unauthorized access.

7. IoT and Embedded Device Forensics

- Covers smart devices, wearables, smart home appliances, industrial systems.
- Complex due to diverse hardware/software.

8. Malware Forensics

- Analyzes malicious software to understand behavior, payload, and impact.
- Uses reverse engineering tools like IDA Pro, Ghidra.

9. Drone and Vehicle Forensics

- Deals with drones, vehicle infotainment systems, GPS logs, black box data.
- Used in accident and crime investigations.

10. Email and Multimedia Forensics

- Focuses on recovering and analyzing emails, audio, video, images.
- Used in cybercrime, fraud, harassment, copyright cases.

Revisiting Computer Forensics

Computer forensics is the scientific examination and analysis of data held on, or retrieved from, computer storage media in such a way that the information can be used as evidence in a court of law.

Computer Forensic Activities

- Computer forensics activities commonly include:
 - the **secure** collection of computer data
 - the **identification** of suspect data
 - the **examination** of suspect data to determine details such as origin and content
 - the **presentation** of computer-based information to courts of law
 - the **application** of a country's laws to computer practice.

The 3 As of Computer Forensics

- The basic methodology consists of the 3 As:
 - Acquire the evidence without altering or damaging the original
 - Authenticate the image
 - Analyze the data without modifying it

Digital Evidence

- Computer forensics is the task of recovering data that users have hidden or deleted and using it as evidence.
- Evidence can be **inculpatory** or **exculpatory**.
- **Inculpatory Evidence:** Evidence tending to incriminate a defendant or indicate their guilt.
- **Exculpatory Evidence:** Evidence tending to exonerate a defendant or helps establish their innocence.

Fit with Information Assurance

- Computer Forensics is part of the incident response (IR) capability
- Forensic “friendly” procedures & processes
- Proper evidence management and handling
- IR is an integral part of Information Assurance

Computer Forensics Versus Other Related Disciplines

- Computer forensics
 - Investigates data that can be retrieved from a computer's hard disk or other storage media
- Network forensics
 - Yields information about how a perpetrator or an attacker gained access to a network
- **Data recovery**
 - Recovering information that was deleted by mistake or lost during a power surge or server crash
 - **Typically you know what you're looking for**
- **Disaster recovery**
 - Uses computer forensics techniques to retrieve information their clients have lost

Computer Forensics Versus Other Related Disciplines (continued)

Investigators often work as a team to make computers and networks secure in an organization

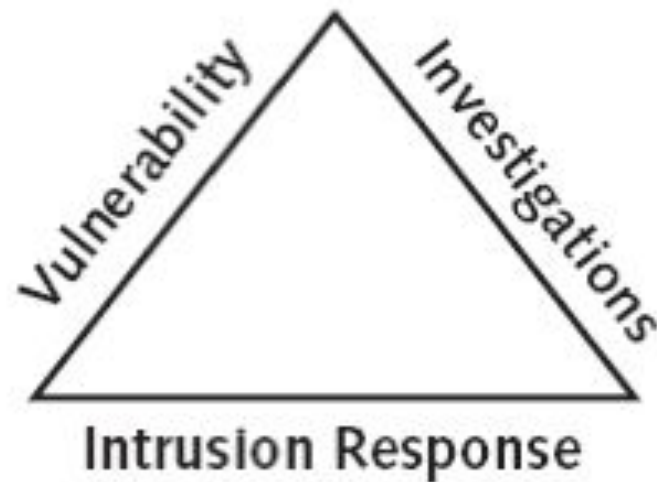


Figure 1-2 The investigations triad

Computer Forensics Versus Other Related Disciplines (continued)

- **Enterprise network environment**
 - Large corporate computing systems that might include disparate or formerly independent systems
- **Vulnerability assessment and risk management group**
 - Tests and verifies the integrity of standalone workstations and network servers
 - Professionals in this group have skills in **network intrusion detection and incident response**

Computer Forensics Versus Other Related Disciplines (continued)

- **Litigation**

- Legal process of proving guilt or innocence in court

- **Computer investigations** group

- Manages investigations and conducts forensic analysis of systems suspected of containing evidence related to an incident or a crime

Crime Scene Investigation

Locard's Exchange Principle

- Locard's Principle states: 'Every contact leaves a trace.'
- Developed by Dr. Edmond Locard.
- Foundation of forensic science.

Origin and History

- Proposed in early 20th century by Edmond Locard.
- Locard established the first forensic laboratory in Lyon, France.
- Inspired modern crime scene investigation.

Core Concept

- When two objects come into contact, material is exchanged.
- Suspect $\leftrightarrow$ Victim $\leftrightarrow$ Crime Scene.
- Evidence may be physical, biological, or trace.

Applications in Forensics

- Used in crime scene analysis.
- Examples: fingerprints, fibers, blood, hair, soil.
- Helps link suspects to victims and crime scenes.

Limitations and Criticisms

- Not all traces are detectable.
- Evidence can be contaminated or lost.
- Relies on careful collection and preservation.

Crime Scene Investigation

- Crime scene investigation is the most crucial step in any investigative process.
- It is the starting point for establishing what has occurred.

Crime Scene

- **Evidence**

“Physical objects that establish that a crime has been committed, can provide a link between a crime and its victim, or can provide a link between a crime and the perpetrator”
(*Saferstein, 2004*).

- ***Digital Evidence***

Digital data that establish that a crime has been committed, can provide a link between a crime and its victim, or can provide a link between a crime and the perpetrator (Carrier & Spafford, 2003)

Crime Scene

- **Crime Scene:**

- “The environment where evidence can potentially exist” (Lee, 2001).
- The environment where the first criminal act occurred is the **PRIMARY** physical crime scene and subsequent scenes are **SECONDARY** physical crime scenes (Lee, 2001).

- **Digital Crime Scene**

- The electronic environment where digital evidence can potentially exist (Rogers, 2005).

Crime Scene Investigative Phases

Recognition



Identification



Individualization



Reconstruction

Documentation

Comparison

Evaluation
Interpretation

Reporting
Presentation

Collection
Preservation

Crime Scene Investigative Phases

- **Recognition**

- knowing what to look for, what constitutes potential evidence, and more importantly what can be ignored. This phase also includes the collection of the evidence.

- **Identification**

- once evidence or potential evidence has been recognized, it needs to be identified. Identification consists of classification at the most basic level based on class characteristics (e.g., hair, blood, fingerprint). This acts as the foundation for the next phases.
- the collected and identified evidence needs to be compared to some standard or control to determine that it came from a particular class (e.g., paint from a 1975 Ford Mustang).

Crime Scene Investigative Phases

- **Individualization**

- the evidence is then further examined to determine any unique characteristics that would allow it to be differentiated from the larger category to a specific person, or entity/object based on its unique characteristics (e.g., paint from a 1975 ford Mustang owned by the primary suspect).

- **Reconstruction**

- the last phase ties together the previous phases and allows the investigator to pull together the pieces of what has been to this point part of puzzle with no real picture to follow, into a logical sequence of events consistent with established timelines.

Digital Crime Scene Phases

- **At Crime Scene**

- Evidence Identification
- Evidence Collection

- **Transportation**

- **In Lab**

- Analysis
- Reporting

Forensic Principles

- Digital/ Electronic evidence is extremely volatile!
- Once the evidence is contaminated it cannot be de-contaminated!
- The court's acceptance is based on the best evidence principle.
- If evidence gets suppressed it is useless
- Chain of Custody is crucial

Forensic Principles

- 5 “Rules” for Evidence
 - Admissible
 - Must be able to be used in court or elsewhere
 - Authentic
 - Evidence relates to incident in relevant way
 - Complete
 - No tunnel vision
 - Reliable
 - No question about authenticity & veracity
 - Believable
 - Clear, easy to understand, and believable by a jury

Forensic Principles

- Chain of Custody
 - Protects integrity of the evidence
 - Effective process of documenting the complete journey of the evidence during the life of the case
 - Allows you to answer the following questions:
 - Who collected it?
 - How & where?
 - Who took possession of it?
 - How was it stored & protected in storage?

Forensic Principles

The IOCE/SWGDE International Principles are governed by the following attributes:

1. Consistency with all legal systems
2. Allowance for the use of a common language
3. Durability
4. Ability to cross international boundaries
5. Ability to instill confidence in the integrity of the evidence
6. Applicability to all forensic evidence
7. Applicability at every level, including that of individual, agency, and country

Forensic Principles

- The 6 Principles are:
 1. When dealing with digital evidence, all of the general forensic and procedural principles must be applied.
 2. **Upon seizing digital evidence, actions taken should not change that evidence.**
 3. When it is necessary for a person to access original digital evidence, that person should be trained for the purpose.
 4. All activity relating to the seizure, access, storage or transfer of digital evidence must be fully documented, preserved and available for review.
 5. An Individual is responsible for all actions taken with respect to digital evidence whilst the digital evidence is in their possession.
 6. Any agency, which is responsible for seizing, accessing, storing or transferring digital evidence is responsible for compliance with these principles.

General Evidence Dos & Don'ts

1. Minimize Handling/Corruption of Original Data
2. Account for Any Changes and Keep Detailed Logs of Your Actions
3. Comply with the Five Rules of Evidence
4. Do Not Exceed Your Knowledge
5. Follow Your Local Security Policy and Obtain Written Permission
6. Capture as Accurate an Image of the System as Possible
7. Be Prepared to Testify
8. Ensure Your Actions are Repeatable
9. Work Fast
10. Proceed From Volatile to Persistent Evidence

Digital Evidence Identification

- Devices that can store Digital Evidence
 - Desktop Computers, Laptops
 - Blackberries, PDAs, Cell phones, Cameras, Video Recorders
 - MP3 Players, IPODS, Xboxes, DVRs, TIVOs
 - Diskettes, CDs, DVDs, USB Drives, Removable Hard Drives, Backup Tapes, Memory Sticks
 - Network Hard Drives, Wireless/Wired Routers, etc.

Device Identification



At Scene Procedures

Step 1: Plan the Seizure

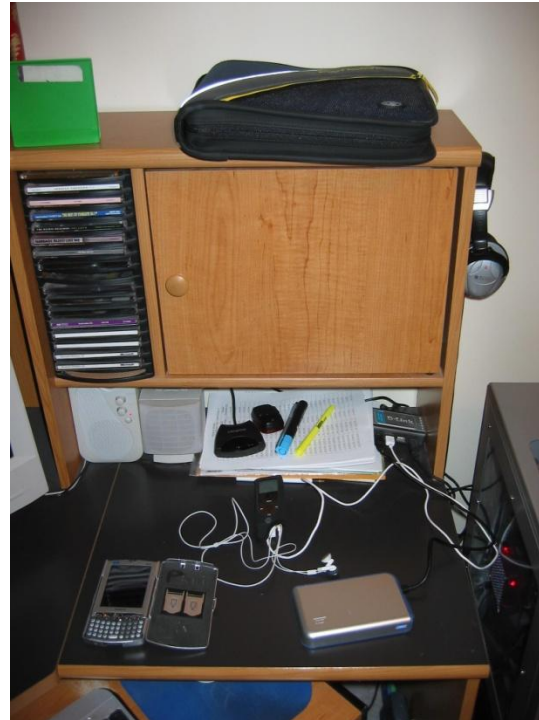
- Crucial — Have a plan in place!
 - Control the scene! — Important!
 - Make sure you have the authority to obtain the evidence
 - What type of case are you investigating?
 - Level of computer knowledge of the suspect?
 - Type of devices that are involved
- Survey the scene
 - Evaluate what you found in the scope of the scene and determine what further assistance you might need for the next phases

At Scene Procedures

Step 2: Documentation

- Document the scene – Important!
 - Make notes about the overall scene. Be sure to include location of evidence and people
 - Take overall general photographs and/or video of the locations of the digital evidence
 - Be sure not to overlook documents near the digital evidence. (post-it notes, printouts, labels) – Seize these documents
 - Take note of any Internet/network devices

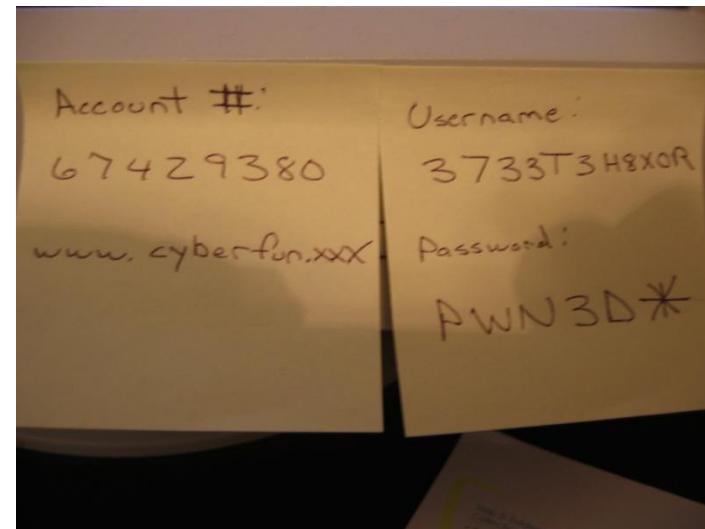
Step 2: Documentation



At Scene Procedures

Step 2: Documentation

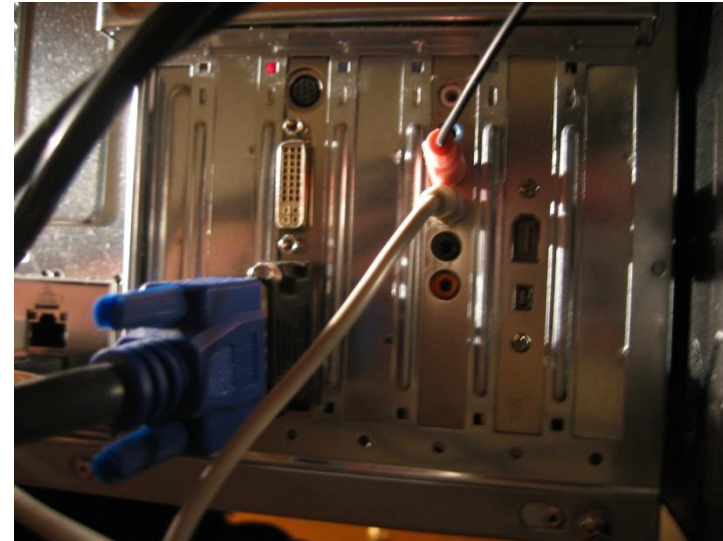
- Take detailed photos and notes of the computer / monitor
 - If the computer is “on”, take photos of what is displayed on the monitor — DO NOT ALTER THE SCENE



At Scene Procedures

Step 2: Documentation

- Make sure to take photos and notes of all connections to the computer/other devices



At Scene Procedures

Step 3: Securing the System

- If the computer is off?
 - Don't turn it on.
- If the computer is on?
 - It depends!
- Disconnect all remote access to the computer (Modems, DSL/Cable Modems, LAN)
 - Note this can be done upon arrival to prevent another party from remotely destroying evidence
- What about WiFi?
- Disconnect all other connections to peripherals (PDAs, Printers, Scanners) and inventory them.

At Scene Procedures

Step 3: Securing the System

- Collect and Inventory all removable media items. (CDs/DVDs, Memory Sticks, Hard Drives, Disks, Backup Tapes, etc..)
- Tag all Items with evidence tags/bags
- Store evidence items in the proper containers with padding to prevent damage
- Place magnetic items in anti-static bags to protect evidence
- Secure the items safely for transport to the station/lab.
- When in doubt... Always Document!!!

At Scene Procedures

Step 4: Interviews

- Separate potential witnesses and suspects to obtain statements
- Ask as much information about the computer system as possible at the scene
 - Username/Passwords/Email Accounts/IMs
 - Who uses the computers?
 - What is the computer used for?
 - Just ask, What illegal things are on the computer and where are they located?
 - Are there any wiping viruses/programs/logic bombs/explosive devices?

At Scene Procedures

Step 5: Transportation

- Transportation and Storage
 - Keep electronics evidence away from magnetic sources (e.g. radio transmitters, speaker magnets and heated seats)
 - Protect evidence from extremes in temperature
 - Use proper anti-shock packing material in containers (i.e., bubble wrap, Styrofoam etc.)
 - Prolonged storage can results in alteration of system evidence (dates, times, etc.) as batteries have a limited life span.
 - Store all seized evidence in a properly secured storage area (e.g., locked cabinet, restricted access lab, etc.)

Forensic Field Kits

- Documentation Tools
 - Cable Tags, evidence tags, felt tip markers
 - Notepad, labels
 - List of phone numbers for technical assistance
 - Digital Camera, Camcorder, extra batteries
- Disassembly and Removal Tools
 - A variety of nonmagnetic sizes and types of:
 - Screwdrivers, Hex and star drivers, Secure-bit drivers
 - Needle nose pliers, standard pliers, tweezers, wire cutters
 - Small flashlight

Forensic Field Kits

- Package/Transport/Other Supplies
 - Anti-static bags, Anti-static bubble wrap
 - Latex, anti-static gloves
 - Cable ties, rubber bands
 - Evidence bags, Evidence tape
 - Packing tape, Masking tap, Box knife
 - Packing material that doesn't produce static
 - Sturdy boxes to store evidence items
 - Hand truck
 - Blank 3.5'' Floppy Disks

Understanding Computer Investigations

Preparing a Computer Investigation

- Role of computer forensics professional is to gather evidence to prove that a suspect committed a crime or violated a company policy
- Collect evidence that can be offered in court or at a corporate inquiry
 - Investigate the suspect's computer
 - Preserve the evidence on a different computer
 - Follow an accepted procedure to prepare a case
- Maintain Chain of custody
 - Route the evidence takes from the time you find it until the case is closed or goes to court

An Overview of a Computer Crime

- Computers can contain information that helps law enforcement determine:
 - Chain of events leading to a crime
 - Evidence that can lead to a conviction
- Law enforcement officers should follow proper procedure when acquiring the evidence
 - Digital evidence can be easily altered by an overeager investigator
- Information on hard disks might be **password protected**

Examining a Computer Crime



Figure 2-1 The crime scene

An Overview of a Company Policy Violation

- Employees misusing resources can cost companies millions of dollars
- Misuse includes:
 - Surfing the Internet
 - Sending personal e-mails
 - Using company computers for personal tasks

Taking a Systematic Approach

- Steps for problem solving
 - Make an initial assessment about the type of case you are investigating
 - Determine a preliminary design or approach to the case
 - Create a detailed checklist
 - Determine the resources you need
 - Obtain and copy an evidence disk drive

Taking a Systematic Approach (continued)

- Steps for problem solving (continued)
 - Identify the risks
 - Mitigate or minimize the risks
 - Test the design
 - Analyze and recover the digital evidence
 - Investigate the data you recover
 - Complete the case report
 - Critique the case

Assessing the Case

- Systematically outline the case details
 - Situation
 - Nature of the case
 - Specifics of the case
 - Type of evidence
 - Operating system
 - Known disk format
 - Location of evidence

Assessing the Case (continued)

- Based on case details, you can determine the case requirements
 - Type of evidence
 - Computer forensics tools
 - Special operating systems

Planning Your Investigation

- A basic investigation plan should include the following activities:
 - Acquire the evidence
 - Complete an evidence form and establish a chain of custody
 - Transport the evidence to a computer forensics lab
 - Secure evidence in an **approved secure container**

Planning Your Investigation (continued)

- A basic investigation plan (continued):
 - Prepare a forensics workstation
 - Obtain the evidence from the secure container
 - Make a forensic copy of the evidence
 - Return the evidence to the secure container
 - Process the copied evidence with computer forensics tools

Planning Your Investigation (continued)

- An **evidence custody form** helps you document what has been done with the original evidence and its forensics copies
- Two types
 - **Single-evidence form**
 - Lists each piece of evidence on a separate page
 - **Multi-evidence form**

Planning Your Investigation (continued)

Corporation X					
Security Investigations					
This form is to be used for one to ten pieces of evidence.					
Case No.:				Investigating Organization:	
Investigator:					
Nature of Case:					
Location where evidence was obtained:					
Description of evidence:		Vendor Name		Model No./Serial No.	
Item #1					
Item #2					
Item #3					
Item #4					
Item #5					
Item #6					
Item #7					
Item #8					
Item #9					
Item #10					
Evidence Recovered by:			Date & Time:		
Evidence Placed in Locker:			Date & Time:		
Item #	Evidence Processed by		Disposition of Evidence		Date/Time
					Page ____ of ____

Figure 2-2 A sample multi-evidence form used in a corporate environment

Planning Your Investigation (continued)

Metropolis Police Bureau High-tech Investigations Unit This form is to be used for only one piece of evidence. Fill out a separate form for each piece of evidence.			
Case No.:		Unit Number:	
Investigator:			
Nature of Case:			
Location where evidence was obtained:			
Item # ID	Description of evidence:	Vendor Name	Model No./Serial No.
Evidence Recovered by:		Date & Time:	
Evidence Placed in Locker:		Date & Time:	
Evidence Processed by	Disposition of Evidence	Date/Time	
		Page ___ of ___	

Figure 2-3 A single-evidence form

Securing Your Evidence

- Use **evidence bags** to secure and catalog the evidence
- Use computer safe products
 - Antistatic bags
 - Antistatic pads
- Use well padded containers
- Use evidence tape to seal all openings
 - Floppy disk or CD drives
 - Power supply electrical cord
- Write your initials on tape to prove that evidence has not been tampered with
- Consider computer specific temperature and humidity ranges

Procedures for Corporate High-Tech Investigations

Develop formal procedures and informal checklists to cover all issues important to high-tech investigations

Employee Termination Cases

- Majority of investigative work for termination cases involves employee abuse of corporate assets
- Internet abuse investigations
 - To conduct an investigation you need:
 - Organization's Internet proxy server logs
 - Suspect computer's IP address
 - Suspect computer's disk drive
 - Your preferred computer forensics analysis tool

Employee Termination Cases (continued)

- Internet abuse investigations (continued)
 - Recommended steps
 - Use standard forensic analysis techniques and procedures
 - Use appropriate tools to extract all Web page URL information
 - Contact the network firewall administrator and request a proxy server log
 - Compare the data recovered from forensic analysis to the proxy server log
 - Continue analyzing the computer's disk drive data

Employee Termination Cases (continued)

- E-mail abuse investigations
 - To conduct an investigation you need:
 - An electronic copy of the offending e-mail that contains message header data
 - If available, e-mail server log records
 - For e-mail systems that store users' messages on a central server, access to the server
 - Access to the computer so that you can perform a forensic analysis on it
 - Your preferred computer forensics analysis tool

Employee Termination Cases (continued)

- E-mail abuse investigations (continued)
 - Recommended steps
 - Use the standard forensic analysis techniques
 - Obtain an electronic copy of the suspect's and victim's e-mail folder or data
 - For Web-based e-mail investigations, use tools such as FTK's Internet Keyword Search option to extract all related e-mail address information
 - Examine header data of all messages of interest to the investigation

Media Leak Investigations

- In the corporate environment, controlling sensitive data can be difficult
- Consider the following for media leak investigations
 - Examine e-mail
 - Examine Internet message boards
 - Examine proxy server logs
 - Examine known suspects' workstations
 - Examine all company telephone records

Media Leak Investigations (consider)

- Steps to take for media leaks
 - Interview management privately
 - To get a list of employees who have direct knowledge of the sensitive data
 - Identify media source that published the information
 - Review company phone records
 - Obtain a list of keywords related to the media leak
 - Perform keyword searches on proxy and e-mail servers

Media Leak Investigations (consider)

- Steps to take for media leaks (continued)
 - Discreetly conduct forensic disk acquisitions and analysis
 - From the forensic disk examinations, analyze all e-mail correspondence
 - And trace any sensitive messages to other people
 - Expand the discreet forensic disk acquisition and analysis
 - Consolidate and review your findings periodically
 - Routinely report findings to management

Industrial Espionage Investigations

- All suspected industrial espionage cases should be treated as criminal investigations
- Staff needed
 - Computing investigator who is responsible for disk forensic examinations
 - Technology specialist who is knowledgeable of the suspected compromised technical data
 - Network specialist who can perform log analysis and set up network sniffers
 - Threat assessment specialist (typically an attorney)

Industrial Espionage Investigations (continued)

- Guidelines
 - Determine whether this investigation involves a possible industrial espionage incident
 - Consult with corporate attorneys and upper management
 - Determine what information is needed to substantiate the allegation
 - Generate a list of keywords for disk forensics and sniffer monitoring
 - List and collect resources for the investigation

Industrial Espionage Investigations (continued)

- Guidelines (continued)
 - Determine goal and scope of the investigation
 - Initiate investigation after approval from management
- Planning considerations
 - Examine all e-mail of suspected employees
 - Search Internet newsgroups or message boards
 - Initiate physical surveillance
 - Examine facility physical access logs for sensitive areas

Industrial Espionage Investigations (continued)

- Planning considerations (continued)
 - Determine suspect location in relation to the vulnerable asset
 - Study the suspect's work habits
 - Collect all incoming and outgoing phone logs
- Steps
 - Gather all personnel assigned to the investigation and brief them on the plan
 - Gather resources to conduct the investigation

Industrial Espionage Investigations (continued)

- Steps (continued)
 - Place surveillance systems
 - Discreetly gather any additional evidence
 - Collect all log data from networks and e-mail servers
 - Report regularly to management and corporate attorneys
 - Review the investigation's scope with management and corporate attorneys

Interviews and Interrogations in High-Tech Investigations

- Becoming a skilled interviewer and interrogator can take many years of experience
- **Interview**
 - Usually conducted to collect information from a witness or suspect
 - About specific facts related to an investigation
- **Interrogation**
 - Trying to get a suspect to confess

Interviews and Interrogations in High-Tech Investigations (continued)

- Role as a computing investigator
 - To instruct the investigator conducting the interview on what questions to ask
 - And what the answers should be
- Ingredients for a successful interview or interrogation
 - Being patient throughout the session
 - Repeating or rephrasing questions to zero in on specific facts from a reluctant witness or suspect
 - Being tenacious